

XLSForm Quality Reviewer

Setup & Introduction Guide

Structural and simulation-based quality assurance for household survey instruments

Standards	World Bank DIME ietestform · IPA ipacheckscto
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Mode	Local installation — data never leaves your machine
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Platforms	macOS · Windows · Linux
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Requires	Python 3.10 or higher
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Introduction

What is XLSForm Quality Reviewer?

XLSForm Quality Reviewer is a locally-run quality assurance tool designed to help researchers, data managers, and survey designers identify structural and logical errors in XLSForm-based household survey instruments before deployment to the field.

The tool performs three complementary types of analysis: a **structural inspection** that examines the form's design without executing it; a **simulation-based analysis** that runs hundreds of synthetic respondents through the form to expose logic errors that structural inspection alone cannot detect; and a **design advisor** that evaluates the form against applied economics and social science survey methodology best practices, returning prioritised improvement suggestions grounded in the XLSForm specification and SurveyCTO documentation. Results are presented in an interactive web interface and can be exported as a formatted HTML report.

Why Quality Assurance Matters

Errors in survey instruments — even minor ones — can have significant consequences for data quality and research validity. Common problems such as misconfigured skip logic, duplicate variable names, missing validation rules, or broken group structures may go undetected during manual review, only surfacing during data collection when corrections are costly or impossible.

XLSForm Quality Reviewer systematically checks for these issues before the form reaches enumerators, reducing the risk of data loss, respondent burden, and the need for costly re-surveys.

Key Features

Structural Analysis

Inspects the form design against World Bank DIME ietestform and IPA ipacheckscto specifications — without executing the form. Detects broken group structures, duplicate field names, missing labels, outdated syntax, and more.

Respondent Path Simulation

Generates realistic synthetic respondents with correlated demographic profiles and response style archetypes (neutral, agreeable, cautious, extreme). Runs them through the form to detect unreachable questions, broken skip chains, and logic conditions that never fire across diverse respondent profiles.

Design Improvement Suggestions

Evaluates the form against applied economics and social science survey methodology best practices. Returns 24 prioritised checks across 13 categories: reference integrity, choice logic, validation bounds, skip logic, performance, multilingual completeness, GPS accuracy, form metadata, and more — grounded in the XLSForm spec and SurveyCTO docs.

Exportable Quality Report

Generates a self-contained HTML report with a metrics dashboard, issue breakdown charts, and grouped issue tables with remediation guidance for each problem type. Suitable for sharing with colleagues or archiving as pre-registration documentation.

Local & Offline Operation

The application runs entirely on the user's machine. No data is transmitted to any external server at any point. Safe for sensitive survey instruments and restricted data.

Severity & Priority Classification

Structural and simulation issues are classified as Critical / High / Medium / Low. Design suggestions carry High / Medium / Low priority ratings. Every finding includes a plain-language explanation and a concrete, copy-ready fix.

Who Is This Tool For?

- **Survey Designers** — Validate XLSForm structure and logic before pilot testing or deployment.
- **Data Managers** — Ensure all field names conform to platform limits and naming conventions, and that metadata fields required for tracking are present.
- **Principal Investigators** — Review a concise quality report to confirm the instrument meets methodological standards prior to field work.
- **Survey Platform Teams** — Identify SurveyCTO or ODK compatibility issues that would cause form upload or rendering failures.

Quality Standards

Structural checks are derived from two widely adopted quality assurance frameworks for household survey instruments. Design suggestions are grounded in the XLSForm specification and SurveyCTO form design documentation.

Source	Institution / Publisher	Scope
ietestform.ado	World Bank DIME Analytics	Field name length, group structure, metadata, outdated syntax, constraint messages, choice list hygiene
ipacheckscto.ado	Innovations for Poverty Action (IPA)	Metadata fields, disabled/read-only flags, required field logic, or_other syntax, integer constraints, duplicate codes
XLSForm Specification	XLSForm.org (ODK / KoBoToolbox)	Field types, column definitions, skip logic syntax, choice list rules, appearance options, repeat groups, multilingual forms
SurveyCTO Form Design Docs	Dobility / SurveyCTO	Performance limits, pulldata() best practices, GPS accuracy parameters, repeat group sizing, calculate field optimisation, form version management

How to Use the Tool

1 Install the application

Follow the platform-specific setup instructions in this guide (Section 2). Installation is required only once; subsequent launches use the provided one-click launcher script.

2 Launch the application

Start the app using the provided launcher script or the terminal command. The tool opens in your web browser at <http://localhost:8501>.

3 Upload your XLSForm

Click the upload area and select your .xlsx file. The tool will immediately parse the survey and choices sheets, detect languages, and display summary statistics.

4 Review structural issues

Structural checks run automatically on upload. Review issues grouped by severity — Critical and High issues should be resolved before proceeding to simulation or field testing.

5 Run respondent simulations

Configure the number of simulations (50–100 recommended) and click Run. The tool generates synthetic respondents with realistic demographic profiles and response styles, then reports unreachable questions and broken skip chains.

6 Review design suggestions

Scroll to the Design Improvement Suggestions section for prioritised best-practice recommendations grounded in the XLSForm specification and SurveyCTO documentation. Each suggestion includes a ready-to-apply fix example.

7 Export the quality report

Download the HTML report for documentation, sharing with colleagues, or archiving as part of the study's pre-registration materials.

Data Privacy Statement

XLSForm Quality Reviewer processes all data locally on the user's machine. No XLSForm content, analysis results, or any other information is transmitted to any external server, cloud service, or third party at any point during use. The tool is safe for use with sensitive survey instruments.

Local Installation Guide

What You Need

Confirm you have received the following files and place them together in the same folder (for example, a folder named *XLSForm Reviewer* on your Desktop).

File	Purpose
app.py	The application
requirements.txt	List of software dependencies
launch_mac.command	One-click launcher for macOS (optional)
launch_windows.bat	One-click launcher for Windows (optional)

macOS Instructions

Step 1 Check Python is installed

Open **Terminal** (+ Space, type *Terminal*, press Enter) and run the command above. You should see **Python 3.10.x** or **3.11.x**.

If Python is not found or below 3.10: download the latest 3.11.x installer from python.org/downloads and follow the prompts.

```
python3 --version
```

Step 2 Navigate to the app folder

Run `ls` to confirm `app.py` and `requirements.txt` are listed.

```
cd ~/Desktop/"XLSForm Reviewer"
```

Step 3 Create a virtual environment (first time only)

Creates an isolated environment so the app's dependencies do not affect other software on your machine.

```
python3 -m venv venv
```

Step 4 Activate the virtual environment

The terminal prompt will change to show `(venv)`, confirming it is active.

```
source venv/bin/activate
```

Step 5 Install dependencies (first time only)

Downloads and installs all required packages. This may take 1–2 minutes.

```
pip install -r requirements.txt
```

Step 6 Launch the application

A browser window will open at `http://localhost:8501`. If it does not open, paste that address into any browser.

```
streamlit run app.py
```

Returning users — quick launch

```
cd ~/Desktop/"XLSForm Reviewer"  
source venv/bin/activate  
streamlit run app.py
```

To **stop the app**, press `Ctrl + C` in Terminal.

Windows Instructions

Step 1 Check Python is installed

Open **Command Prompt** (Win + R, type `cmd`, Enter). You should see **Python 3.10.x** or **3.11.x**.

If not found or below 3.10: download the 3.11.x 64-bit installer from python.org/downloads/windows. On the first screen, tick **"Add Python to PATH"** before clicking Install Now.

```
python --version
```

Step 2 Navigate to the app folder

Run `dir` to confirm the files are listed.

```
cd %USERPROFILE%\Desktop\XLSForm Reviewer
```

Step 3 Create a virtual environment (first time only)

```
python -m venv venv
```

Step 4 Activate the virtual environment

The prompt will show (venv).

```
venv\Scripts\activate
```

*If activation is blocked by an execution policy error, open PowerShell as Administrator and run:
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser Then return to Command Prompt.*

Step 5 Install dependencies (first time only)

```
pip install -r requirements.txt
```

Step 6 Launch the application

A browser window will open at <http://localhost:8501>.

```
streamlit run app.py
```

Returning users — quick launch

```
cd %USERPROFILE%\Desktop\XLSForm Reviewer  
venv\Scripts\activate  
streamlit run app.py
```

To **stop the app**, press **Ctrl + C** in Command Prompt.

Linux Instructions (Ubuntu / Debian)

Step 1 Check Python is installed

Open a terminal and run the command above. **If Python 3.10 or higher is not available**, install it:

```
python3 --version
```

```
sudo apt update
```

```
sudo apt install python3 python3-pip python3-venv -y
```

Step 2 Navigate to the app folder

Run **ls** to confirm the files are present.

```
cd ~/Desktop/"XLSForm Reviewer"
```

Step 3 Create a virtual environment (first time only)

```
python3 -m venv venv
```

Step 4 Activate the virtual environment

The prompt will show (venv).

```
source venv/bin/activate
```

Step 5 Install dependencies (first time only)

```
pip install -r requirements.txt
```

Step 6 Launch the application

The app will open at <http://localhost:8501>. Navigate there manually if the browser does not open automatically.

```
streamlit run app.py
```

Returning users — quick launch

```
cd ~/Desktop/"XLSForm Reviewer"  
source venv/bin/activate  
streamlit run app.py
```

To **stop the app**, press **Ctrl + C** in the terminal.

Troubleshooting

Problem	Likely Cause	Solution
python3: command not found	Python is not installed	Follow Step 1 for your OS
pip: command not found	pip not on PATH	Use <code>python3 -m pip install -r requirements.txt</code>
ModuleNotFoundError: No module named 'streamlit'	Virtual environment not active	Run the activate command (Step 4) before launching
Browser does not open	Streamlit cannot detect browser	Navigate manually to <code>http://localhost:8501</code>
Port 8501 already in use	Another instance is running	Stop it with <code>Ctrl + C</code> , or append <code>--server.port 8502</code>
Windows: activate execution policy error	PowerShell security setting	See the PowerShell note in Windows Step 4

Problem	Likely Cause	Solution
openpyxl error on upload	Missing dependency	Re-run Step 5 with the virtual environment active
App shows 0 questions	Incorrect sheet structure	File must have a survey sheet with type and name columns

Frequently Asked Questions

Is my data safe?

Yes. The application runs entirely on your local machine. No XLSForm data, results, or personal information is transmitted to any external server at any point.

Can I run this without an internet connection?

Yes. Once the dependencies are installed (Step 5), the app operates fully offline. An internet connection is only required for the initial installation.

Can multiple people use the same installation?

The app runs on a single local port. For independent concurrent use, each user should install the tool separately in their own folder.

How do I update the app when a new version is provided?

Replace app.py with the new file. If a new requirements.txt is provided, re-run pip install -r requirements.txt with the virtual environment active.

Where are downloaded HTML reports saved?

Reports are saved to your browser's default download folder.

What forms does this tool support?

Any XLSForm-compatible .xlsx file with a survey sheet and type/name columns. The tool is optimised for SurveyCTO and ODK Collect forms.